Numerical Experiments

$$E(y) := \underbrace{\frac{-1}{2}\int_\Omega |\nabla^2 v|^2 - \nu[v,v]}_{\text{membrane}} + \underbrace{\frac{1}{2}c_\nu \int_\Omega |\Delta w|^2 - (1-\nu)[w,w]}_{\text{bending}} + \underbrace{\frac{1}{2}\int_\Omega J^T \nabla^2 v J : \nabla w \otimes \nabla w}_{\text{coupling}} - \gamma\beta^4 \underbrace{\int_\Omega}_{\text{transv. p}}$$

1. Numerical Setup

1.1 Variational Problem

- Brief description of the problem being solved.
- Reference to the finite element discretisation and solver configuration.

1.2 Implementation Details

The numerical solution of the variational problem presented in Section 5 is carried out leveraging the dolfinx/FEniCSx finite element platform, relying on the PETSc library for efficient linear and nonlinear algebra. Our implementation, https://github.com/kumiori/disclinations features high-order polynomial Lagrange finite elements whose order can be changed at runtime, for flexible accuracy in the discretisation of the variational principle. More specifically, we use the Newton nonlinear solver provided by PETSc SNES and without step adjustment ('basic lineasearch', cf. manual). The iterative subproblems are solved by Krylov subspace method (preonly) with LU factorization as preconditioner (pc_type: lu).

The numeircal • Stopping tolerance: snes_stol = 1e-6 • Absolute tolerance: snes_atol = 1e-6 • Relative tolerance: snes_rtol = 1e-6 • Maximum iterations: snes_max_it = 200

Geometry and Mesh

All presented numerical experiments are performed on a circular domain $\Omega := B(0,1)$ (the unit open ball) centered in the origin, meshed with triangular (Delaunay) elements of typical size $\eta = 0.05$. This geometry forms the basis for all parametric verification computations and numerical experiments. The finite element discretisation employs third-order Lagrange elements for both the transverse displacement (w) and the Airy stress potential (v), ensuring sufficient smoothness, allowing for the computation of the Gaussian curvature, whilst capturing the nonlinear coupling between in-plane and out-of-plane elastic behaviour.

The numerical stabilisation coefficient α is fixed to ...

1,595 nodes, xxxx elements, matrix sizes ~ ... K entries

The non-dimensional formulation involves two tunable parameters The aspect ratio ($\beta = R/h \gg 1$), measuring the plate's slenderness and the relative transverse pressure (p_0/E), ratio of applied pressure to material stiffness. Finally, the disclination strength, characterised by the Frank angle for kinematically incompatible loads, is fixed to ± 1 in the following experiments, to qualitatively distinguish positive from negative induced membrane stresses, respectively corresponding to compression and tension.

All computations refer to the non-dimensional formulation.

Nontrivial kinematically incompatible loads, verified against closed-form analytical solutions.

Mixed loading conditions combining surface pressure and disclinations.

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2. Verification Tests

Validation experiments provide substantiation that the computational model accurately represents the mathematical model of the mechanical system within specified limits of accuracy. These experiments are functional in establishing the reliability of the numerical framework and its ability to predict key physical behaviors. The aim of validation is twofold:

1. Quantitative Accuracy: Demonstrating that the numerical results match known solutions within acceptable bounds.
2. Qualitative Insights: Ensuring that the model captures the essential physical mechanisms and trends inherent to the system.

Validation experiments focus on two configurations: kinematically compatible loads through surface pressure distributed over the plate and kinematically incompatible loads with a dipole of disclinations, symmetrically distributed around the centre of the plate.

$\rho(x) = \sqrt{x_1^2 + x_2^2}$

2.1 Plate with Kinematically Compatible Loads: transverse pressure

rescaling nondimensionalisation, all test collapse to ...

Let $\theta = 0$ and $p(x) = p_0 \frac{40}{3}(1-\rho^2(x))^4 + \frac{16}{3}(11+\rho^2(x))$, then \$w(x) \$ = $w_0(1-\rho^2(x))^2$ and $v(x) = v_0(-\frac{1}{12}(1-\rho^2(x))^2 - \frac{1}{18}(1-\rho^2(x))^3 - \frac{1}{24}(1-\rho^2(x))^4)$ solves eq ....

Here, $p_0 = \sqrt{\frac{2D^3}{Eh}}$, $w_0\sqrt{\frac{2D}{Eh}}$, $v_0 = \frac{D}{h}$ In Figures **??** we compare the approximate solutions of the three finite element formulations introduced in section **??** to the analytical solution presented above.

Due to the rescaling, the nondimensional formulation involving only transverse load is independent of the aspect ratio,

the dependence on the aspect ratio is explicit in the rescaling factors for the fields and the energy

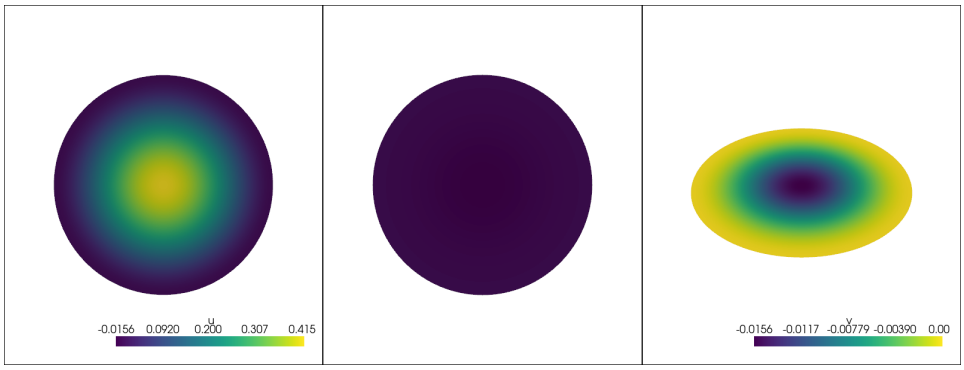
figure ... we compare the profile of the finite element approximate fields with the closed form analytic solution, along the segment $x_2 = 0$

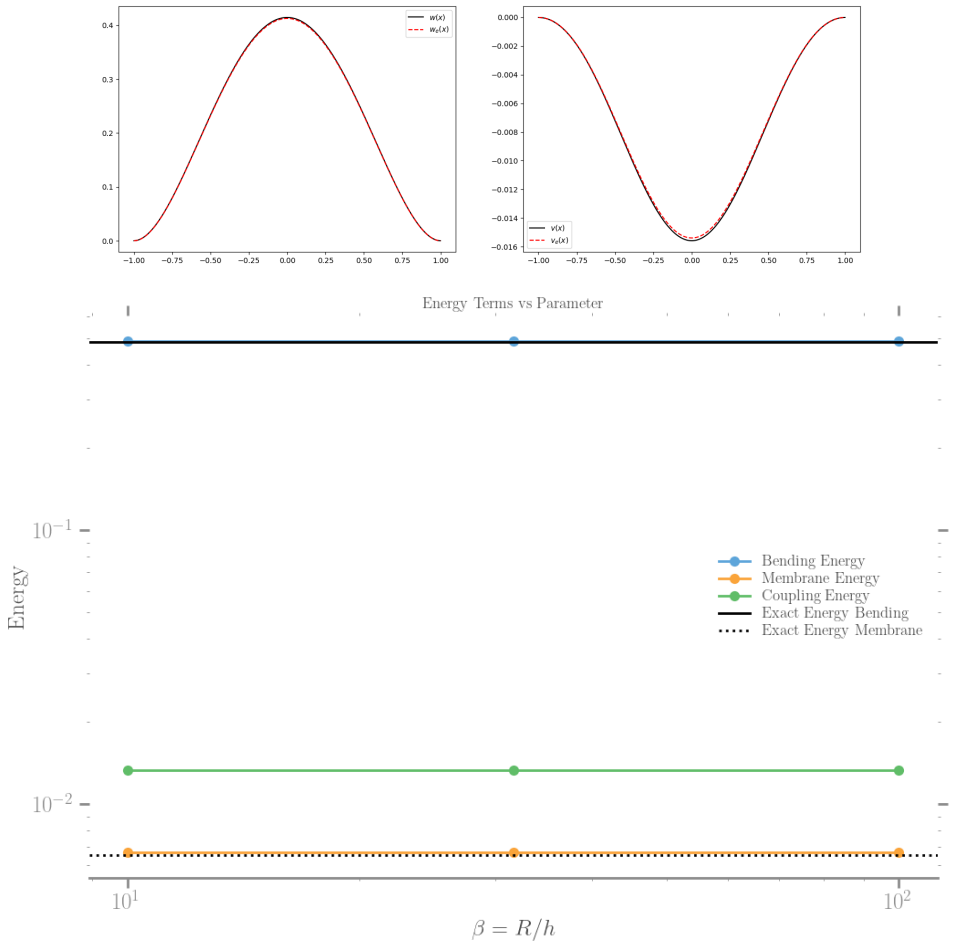
Owing to the interpretation of v as membrane stress potential, its sign locally informs of the positive vs. negative character of membrane strains.

Transverse pressure induces bending which in turn, induces tensile (positive) membrane strains.

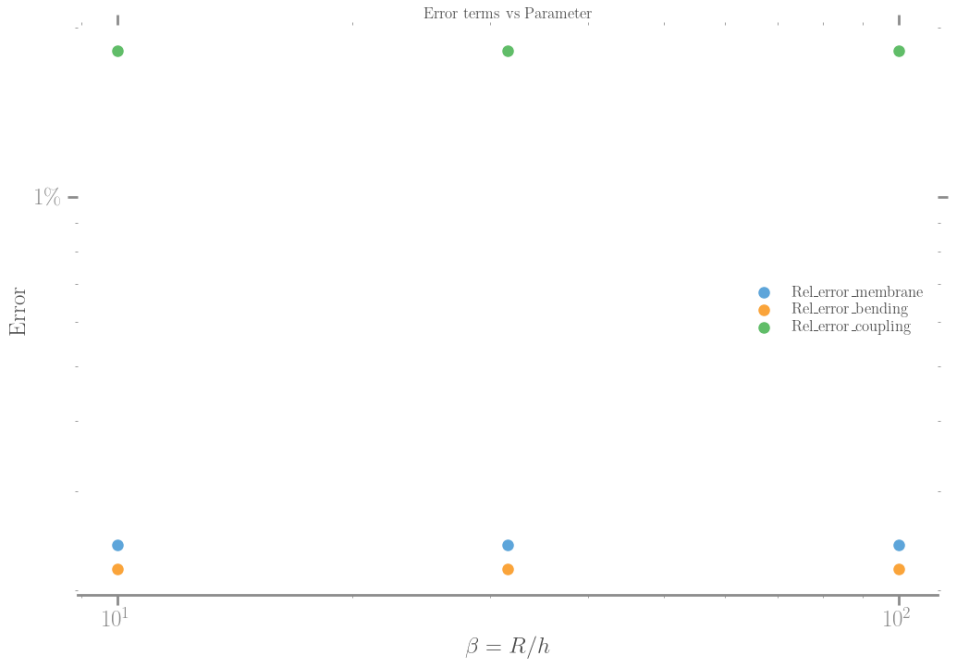
Finally, figure ... shows the Gaussian curvature, computed postprocessing the equilibrium solution, as the determinant of the Hessian of the transverse displacement. Note the inversion of curvature due to the clamping boundary condition. In the image, the red locus corresponds to $x : \kappa(x) = 0$

- **Setup**: Plate loaded with surface pressure.
- **Objectives**: Compare numerical results against known analytic solutions.
- **Results**: Error analysis, visualisations.

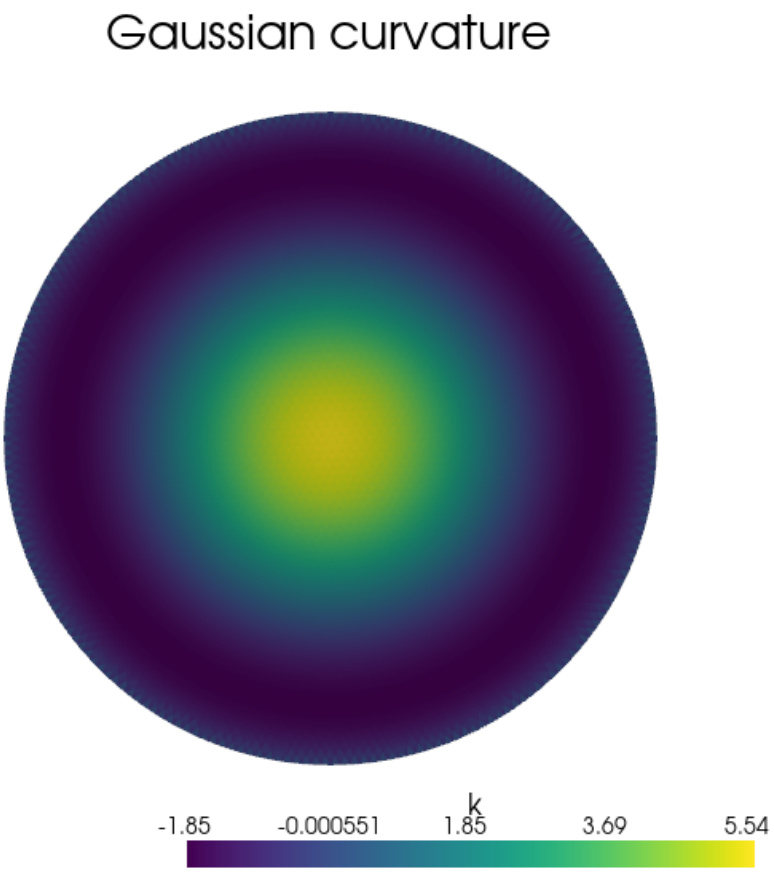


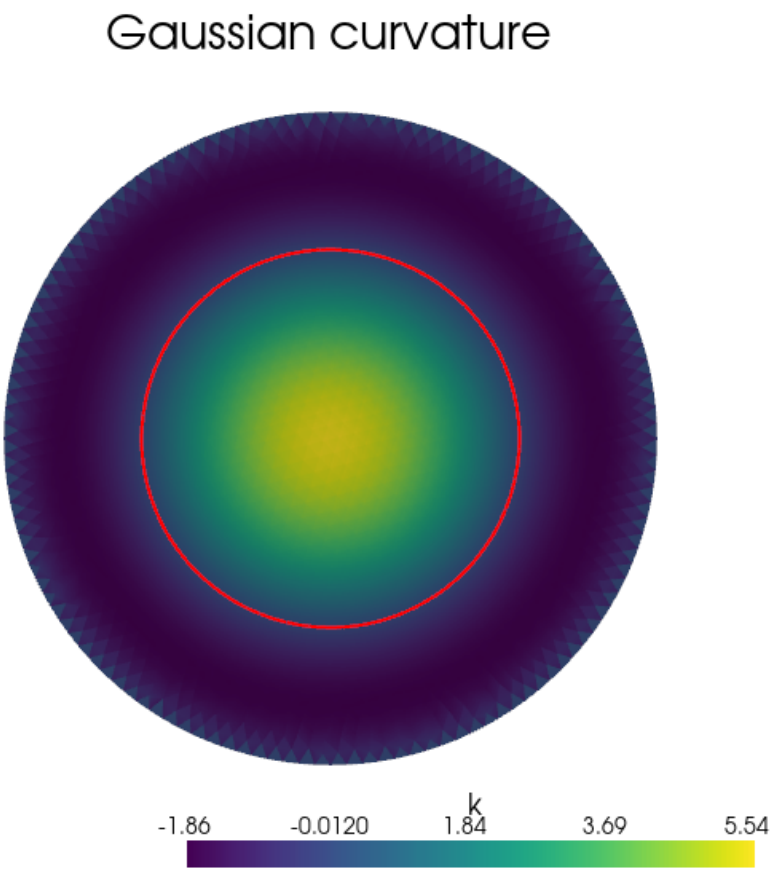


Caption: remove figure, energy (bending, membrane, and coupling) wrt closed



form computation


Gaussian curvature
k
-1.85
-0.000551
1.85
3.69
5.54



Caption: Gaussian curvature, locus of inversion of the curvature Extra: compare errors varying penalisation parameter?

2.2 Plate with Kinematically Incompatible Loads, Dipole, Parametric varying aspect ratio

To verify our numerical implementation for a nontrivial incompatible loading against closed form analytic solutions, the second setup involves a plate loaded with a dipole of disclinations symmetrically arranged around the centre of the plate.

Let $\theta(x) = \delta(x - 0.2e_1) - \delta(x + 0.2e_1), p(x) = 0$. Then $w(x) \equiv 0$ and $v(x) = v_0(\rho^2(x)\ln(\rho^2(x)) - \rho^2(x) + 1)$ solves (**??**). here $v_0 = \frac{Ehs}{16\pi}$ We solve $[\dots]$ with $\gamma = 0$, for varying β between 10 and $X00$.

$w = 0$ is an admissible solution for the dipole problem

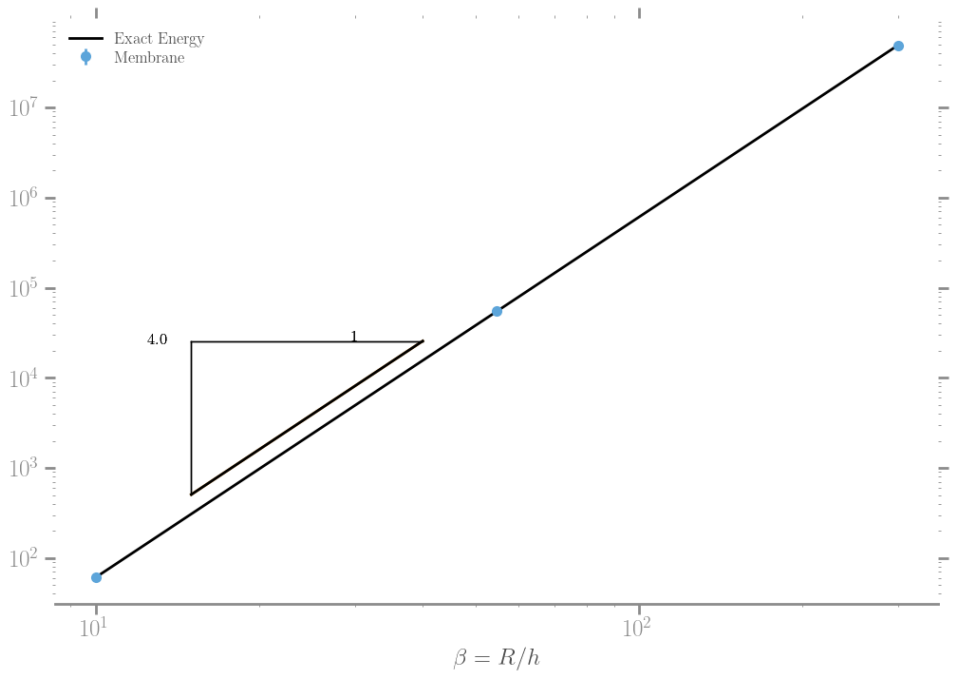
Figure, ..., show energy scaling as a function of the aspect ratio, fourth power of β. This behaviour corresponds to the expected energy scaling, by remarking that the multiplicative factor which scales the membrane equations - linear when $w = 0$ - is β^2. The stress solution is hence quadratic in β^2 thus exponent of the energy. On the other hand, the bending energy and the coupling energy are zero.

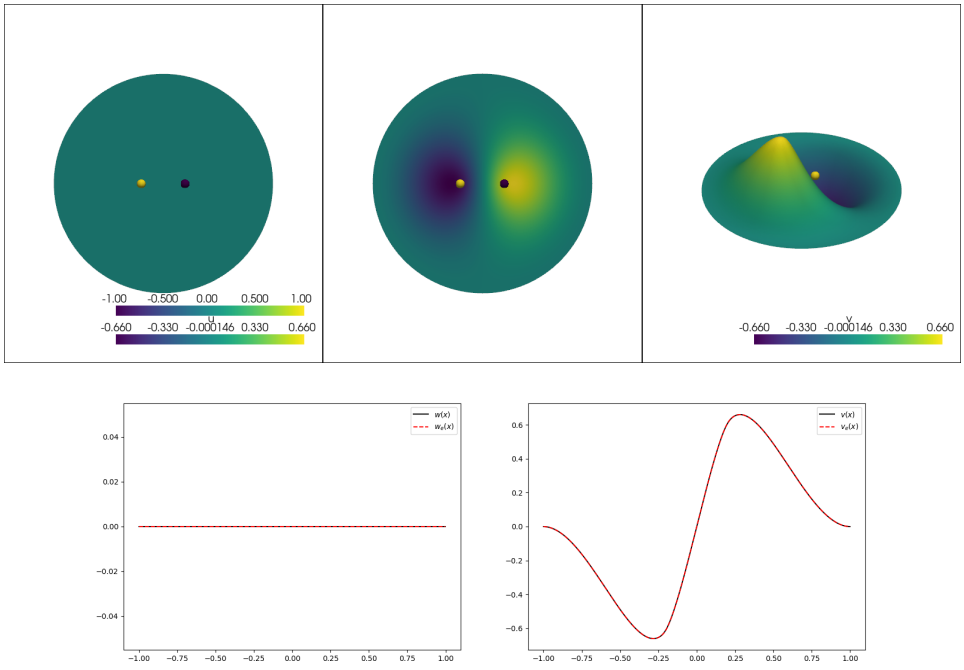
Figure ... compares the profiles of approximate finite element solutions with respect to the closed form analytic solutions.

Figure ... shows the fields w, v on the two-dimensional computational domain, the dots mark the locations of the disclinations (colour coding),

Figure shows the scalar stress potential associated to the positive (colour) and negative (colour) disclination, respectively inducing a locally tensile and compressive state. RK: colours are inverted, FIX

- **Results**: Error analysis, visualisations.





3. Parametric Experiments

3.1 Mixed Loading: Disclination monopole and surface pressure, Varying Thickness

We perform a series of parametric experiments investigating mixed-mode loading combining surface pressure and a disclination monopole, as the plate's aspect ratio is varied whilst keeping the prefactor of the transverse external load $\gamma\beta^4$ constant and equal to one. This implies that, as β varies between 10 and $X00$, the physical dimensional surface pressure decreases with the inverse fourth power of the dimensional thickness.

Nondimensional energy plots show the energy scaling of the different components of the energy. Similarly to the dipole setting, the membrane energy scales with the fourth power of the aspect ratio, whereas the non-dimensional bending energy decreases with a (negative) exponent smaller than -2 [show the bound] (more precisely, -1.65).

For large values of the aspect ratio, small numerical oscillations appear on the bending term, whose order of magnitude is 8 orders of magnitude smaller

than the dominant membrane energy term.

In the dimensionless formulation presented [X], β acts as a gain for the distribution of disclinations, while γ modulates the external load.

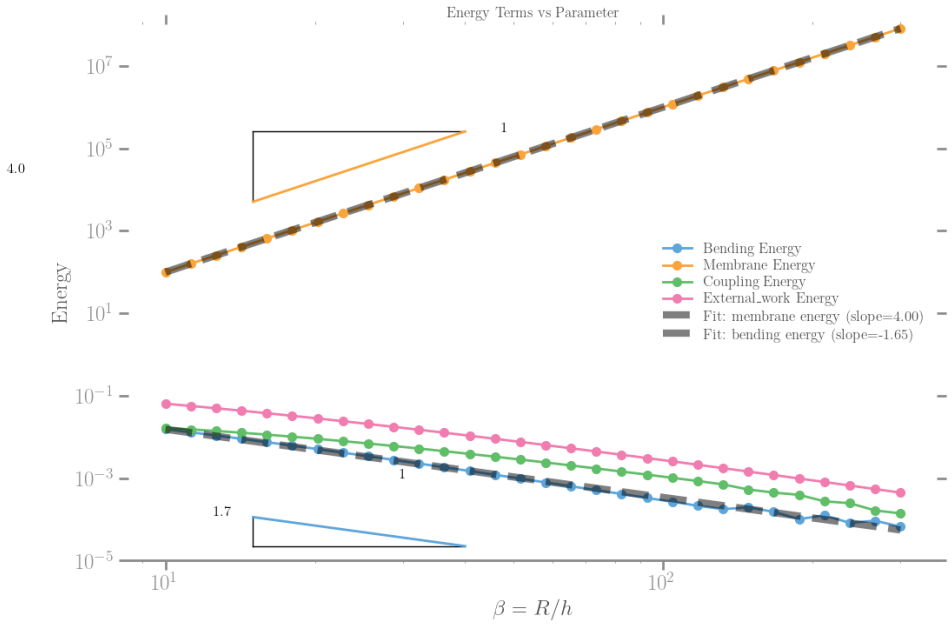
In this section we analyse the mechanics of the system by varying first β in some appropriate range while keeping γ constant; and then varying γ with constant β. In order to do this, we provide an estimate for a realistic range of values for both the dimensionless parameters.

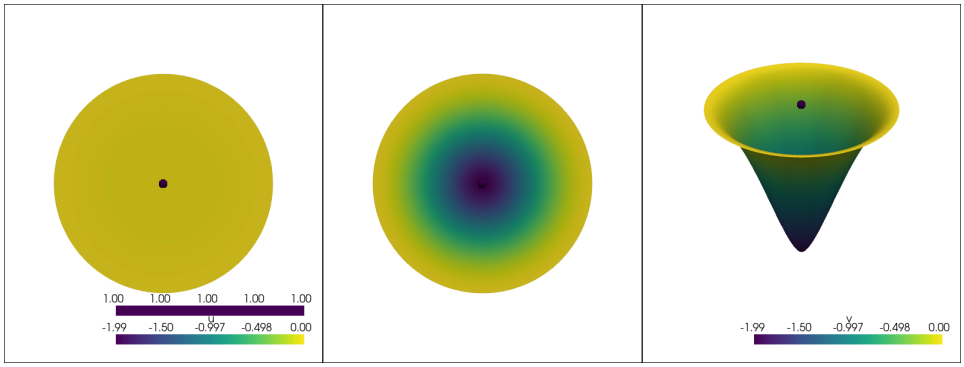
Non-dimensional transverse displacement goes to zero.

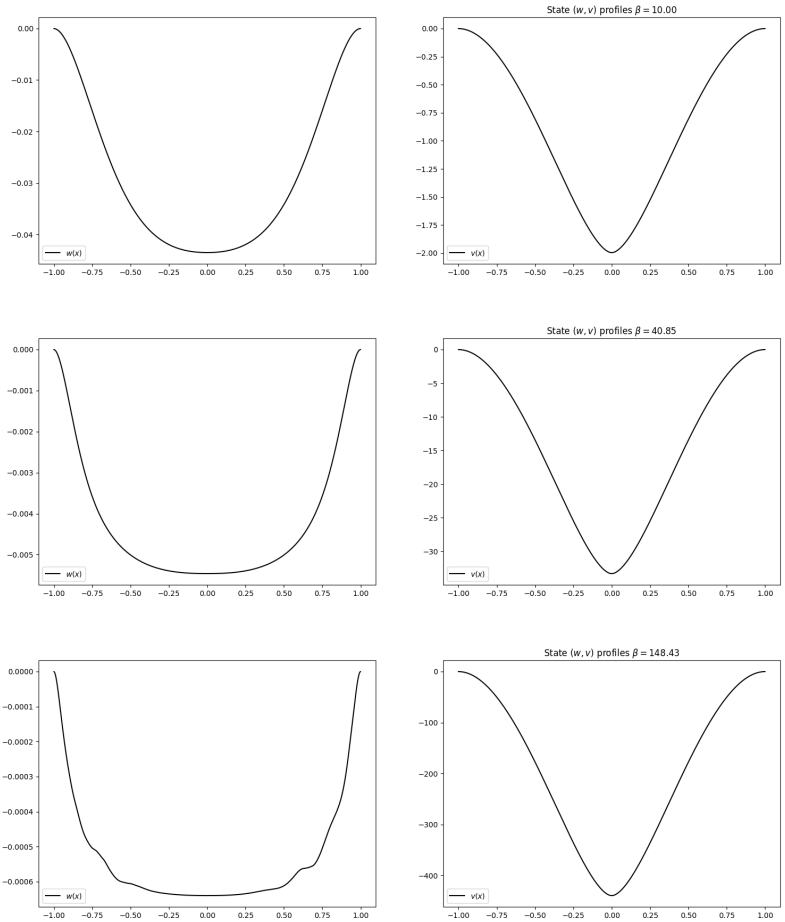
We tune $\overline{p_0}$ to (a multiplier for) the weight of the plate, i.e. considering a density $\rho_{\mathrm{D}} = O(10^3 kg/m^3)$ (corresponding to a reference density for steel/metal) and denoting by g the gravitational acceleration we obtain

$\overline{p_0} \sim \rho_{\mathrm{D}} g h \sim 10^2 \mathrm{Pa}$

- **Results**: Parametric plots (e.g., energy terms, deformation profiles).







For large values of a, h is small and p_0 growths with the fourth power of h. For $p_0 \equiv 0$, $w \equiv 0$ in $B_1(0)$ solves the out-of-plane equation. This is the

reason why in Figure **??** the transverse displacement approaches zero as a increases. As a consequence, v in equation (**??**) is only activated by $\delta(0)$ and hence asymptotically approaches $v^0(x)$, solution to the problem

$$\begin{cases} \Delta^2 v^0 = XXX\delta(0) & \text{in } B_1(0) \\ v^0 = \partial_n v^0 = 0 & \text{on } \partial B_1(0) \end{cases}$$

$$v^0(x) = Ehs\mathcal{G}_\Omega(x) \quad \Rightarrow \|v^0\|_{L^2(\Omega)} = \frac{Ehs}{30\pi} \approx 10^5 N/m$$

where $\mathcal{G}_\Omega$ is the Green function of the biharmonic problem in $H_0^2(\Omega)$ and is equal to

$$\mathcal{G}_\Omega(x) = \frac{1}{16\pi}\Big(1 - |x|^2 + |x|^2 \ln(|x|^2)\Big), \qquad x \in \Omega$$

3.2 Mixed Loading: Varying Pressure

parametric experiment, varying surface pressure in a range that spans 0.1 to 100 times the plate's weight, assuming ... density

relative interplay of bending vs. membrane deformation, changing external loading conditions whilst keeping the aspect ratio and the power of the disclination constant.

interpreted as .. multiplier of plate's weight

Energy plot shows scaling properties of the energy terms

computations for aspect ratio $\beta = 100$

both the disclination and the surface pressure induce same sign curvature this setup shows a relatively weak coupling across the entire range of the loading parameter, the membrane energy is constant and equals the energy estimate provided by the closed form solution for the fundamental monopole solution. On the other hand, the bending energy is quadratic with respect to the load multiplier, as expected by inspecting the (strong form of the) bending problem whilst considering the stress potential a given. As the value of the surface pressure varies, the stress potential shows a self-similar profile (cf images), whereas the transverse displacement progressively localises the deformation around the (clamped) boundary.

- **Setup**: Surface pressure combined with disclination monopole; pressure is varied to span a range ...
- **Results**: Parametric plots (e.g., energy terms, deformation profiles).

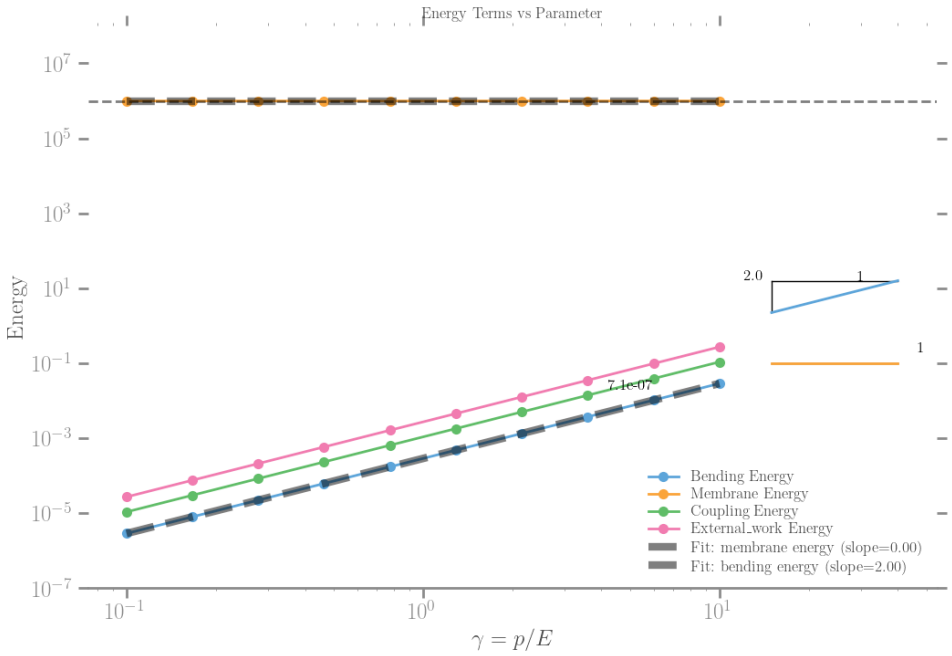
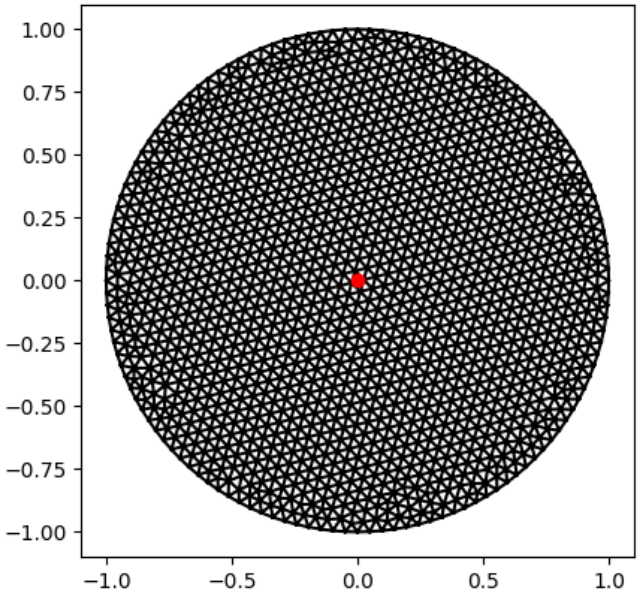
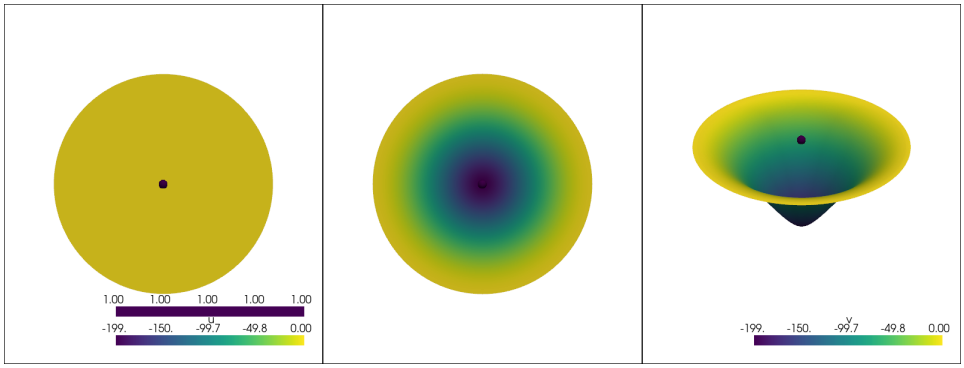
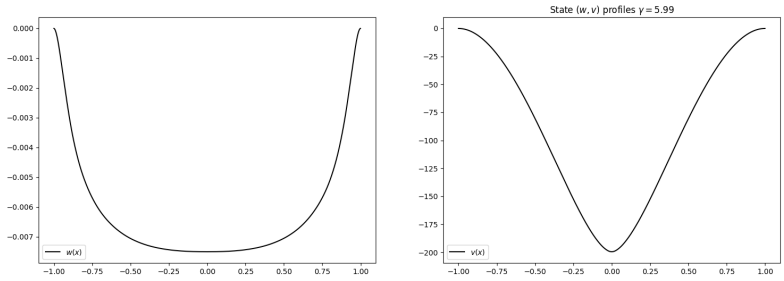


Figure 1: image



typical mesh (with one indicated disclination at the centre)





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4. Complex Experiment: Flower Configuration

4.1 Setup

- **Configuration**: A flower-like arrangement of disclinations mimicking engineered graphene sheets.
- **Objectives**: Demonstrate the capability of the model for complex and nonstandard configurations.

4.2 Results

- Energy analysis, deformation profiles.
- Discussion of insights from this experiment.

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5. Summary of Numerical Results

- Recap key findings from the verification and parametric experiments.
- Discuss implications for the mechanics of thin plates under complex loading scenarios.

Details

- Software environment: dolfinx/FEniCSx, PETSc.
- Solver configurations (e.g., SNES settings, tolerance). solvers: nonlinear: prefix: fvk snes: snes_type: newtonls snes_stol: 1e-6 snes_atol: 1e-6 snes_rtol: 1e-6 snes_max_it: 200 # snes_divergence_tolerance: -1.0 snes_monitor: ” ksp: ksp_type: preonly pc_type: lu

- Meshes geom_type: ‘circle’ radius: 1. mesh_size: 0.05

1. Variational Problem: • What is the mathematical formulation? governing equations, detaild in sec. 5 boundary conditions, H20

• What are the primary unknowns (e.g., displacements, rotations)? transverse displacement w, airy membrane stress potential v • What are the physical parameters involved? Youngs modulus E, thickness h non-dimensional formulation highlights the role of two non-dimensional parameters: aspect ratio, beta, R/h relative transverse pressure, p0/E disclination strength, frank angle (taken either +1 or -1) 2. Finite Element Model: • What is the type of elements used (e.g., Argyris, Lagrange)? FEM lagrange, order: 3 • What is the resolution of the mesh (e.g., element size, refinement levels)?

3. Solver Details: • Which solver (e.g., PETSc’s SNES) and preconditioning methods are used? petsc snes, the linear iterative solver is based on the krylov subspace method, preconditioner is an lu factorisation

• What are the tolerances for convergence (e.g., relative, absolute residuals)? solvers: nonlinear: prefix: fvk snes: snes_type: newtonls snes_stol: 1e-6 snes_atol: 1e-6 snes_rtol: 1e-6 snes_max_it: 200 # snes_divergence_tolerance: -1.0 snes_monitor: ” ksp: ksp_type: preonly pc_type: lu

• Any specific solver settings or configurations? 4. Physical Parameters: • Geometry (e.g., radius, thickness). • Material properties (e.g., elastic moduli). • Loading conditions (e.g., surface pressure,).